

1 (a)	temperature current (allow amount of substance, luminous intensity)	B1	B1	[2]
(b) (i) 1.	$E = (\text{stress/strain}) = [\text{force/area}] / [\text{extension/original length}]$			
	units of stress: $\text{kg m s}^{-2} / \text{m}^2$ and no units for strain	B1		
	units of E : $\text{kg m}^{-1} \text{s}^{-2}$	A0	[1]	
2.	units for T : s, l : m and M : kg			
	$K^2 = T^2 E / M l^3$ hence units: $\text{s}^2 \text{kg m}^{-1} \text{s}^{-2} / \text{kg}^3 (= \text{m}^{-4})$	C1		
	units of K : m^{-2}	A1	[2]	
(ii)	% uncertainty in $E = 4\%$ (for T^2) + 0.6% (for l^3) + 0.1% (for M) + 3% (for K^2) $= 7.7\%$	B1		
	$E = [(1.48 \times 10^5)^2 \times 0.2068 \times (0.892)^3] / (0.45)^2$ $= 1.588 \times 10^{10}$	C1		
	7.7% of $E = 1.22 \times 10^9$	C1		
	$E = (1.6 \pm 0.1) \times 10^{10} \text{kg m}^{-1} \text{s}^{-2}$	A1	[4]	